Data types

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Common data types

- Text data types
  - `CHAR`, `VARCHAR` and `TEXT`
- Numeric data types
  - `INT` and `DECIMAL`
- Date / time data types
  - `DATE`, `TIME`, `TIMESTAMP`, `INTERVAL`
- Arrays

Determining data types from existing tables

```sql
SELECT
    column_name,
    data_type
FROM INFORMATION_SCHEMA.COLUMNS
WHERE column_name in ('title','description','special_features')
  AND table_name ='film';
```

```
+-----------------+------------------+
| column_name     | data_type        |
|-----------------|------------------|
| title           | character varying |
| description     | text             |
| special_features | ARRAY            |
+-----------------+------------------+
```

We can do INFORMATION_SCHEMA.TABLES too

Dates

TIMESTAMP data types

- **ISO 8601 format: yyyy-mm-dd**

```
+-----------------------------+
| timestamp                   |
|-----------------------------|
| 2019-03-26 01:05:17.93027+00 |
+-----------------------------+
```

DATE and TIME data types

```
+------------+------------------+
| date       | time             |
|------------|------------------|
| 2005-05-28 | 01:05:17.93027+00 |
+------------+------------------+
```

TIME data type is stored without a timezone by default

INTERVAL data types

```
+-----------+
| interval  |
|-----------|
| 4 days    |
+-----------+
```

```
SELECT rental_date + INTERVAL '3 days' as expected_return
FROM rental;
```

```
+--------------------+
| expected_return    |
|--------------------|
| 2005-05-27 22:53:30 |
+--------------------+
```

Useful when we calculate arithmetic

Arrays

CREATE TABLE example

```
CREATE TABLE my_first_table (
    first_column text,
    second_column integer
);
```

INSERT example

```
INSERT INTO my_first_table
    (first_column, second_column) VALUES ('text value', 12);
```

ARRAY a special type

Let's create a simple table with two array columns.

```
CREATE TABLE grades (
    student_id int,
    email text[][],
    test_scores int[]
);
```

INSERT statements with ARRAYS

Example INSERT statement:

```
INSERT INTO grades
    VALUES (1,
    '{{"work","work1@datacamp.com"},{"other","other1@datacamp.com"}}',
```

```
    '{92,85,96,88}' );
```

Accessing ARRAYs

```
SELECT
   email[1][1] AS type,
   email[1][2] AS address,
   test_scores[1],
FROM grades;
```

```
+--------+--------------------+-------------+
| type   |  address           | test_scores |
|--------|--------------------|-------------|
| work   | work1@datacamp.com | 92          |
| work   | work2@datacamp.com | 76          |
+--------+--------------------+-------------+
```

Note that PostgreSQL array indexes start with one and not zero.

Can be used in a WHERE clause to filter as well

ARRAY functions and operators

```
SELECT
   email[2][1] as type,
   email[2][2] as address,
   test_scores[1]
FROM grades
WHERE 'other' = ANY (email);
```

```
+---------+---------------------+-------------+
| type    |  address            | test_scores |
|---------|-----------------------------------|
| other   | other1@datacamp.com | 92          |
| null    | null                | 76          |
+---------+---------------------+-------------+
```

ARRAY functions and operators

```
SELECT
   email[2][1] as type,
   email[2][2] as address,
   test_scores[1]
FROM grades
WHERE email @> ARRAY['other'];
```

```
+---------+---------------------+-------------+
| type    |  address            | test_scores |
|---------|-----------------------------------|
| other   | other1@datacamp.com | 92          |
| null    | null                | 76          |
+---------+---------------------+-------------+
```